Verdure Trace *AYII+* Product Design Note

1. Executive Summary

This note sets out a revised design for a **Verdure Trace Area Yield Index Insurance-plus (*AYII+*)** product that combines **weather data, NDVI-based vegetation monitoring, historical yield evidence, and modelled area-yield estimation** to produce an insurance structure that is independent, transparent, commercially usable, and technically defensible.

The product is designed for contexts where client-provided yield data may be incomplete, optimistic, inconsistent, or difficult to verify independently. The design therefore does not rely blindly on declared yields. Instead, it uses a structured evidence hierarchy in which:

- historical yield remains the core reference where credible,
- NDVI is used to validate, constrain, and extend yield assumptions,
- weather defines peril occurrence and supports burn-cost simulation,
- final payout remains anchored to benchmark yield versus observed or modelled area yield,
- and an optional **cumulative in-season payout layer** can be used to provide staged liquidity before final end-of-season reconciliation.

This is consistent with the AYII reference framework, which requires yield data for benchmark construction and pricing, uses weather for probable maximum loss and catastrophe understanding, and recognizes remote sensing data such as NDVI, EVI, VCI, flood, or drought extent as useful for yield validation and dispute-resolution support. The reference document states that a minimum of 10 years of yield data is required for product design, that weather data is used for product development and PML analysis, and that remote sensing data is used for validation of yield data and understanding data variation.

The fully revised Verdure Trace design improves traditional AYII by introducing:

- an independent yield credibility framework,
- a hybrid weather-plus-NDVI underwriting logic,
- modelled area-yield estimation for stronger final settlement,
- and a cumulative stage-based payout approach that reflects the increasing value at risk as the crop grows.

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2. Product Objective

The proposed product aims to:

1. protect farmers, lenders, aggregators, and insurers against systemic seasonal yield losses at area level,
2. reduce dependence on unverified client-entered yield data,

3. use objective data sources to determine insurable yield levels,
4. support premium adequacy through burn-cost simulation,
5. enable in-season crop monitoring and early payout indication,
6. fit naturally into the Verdure Trace platform architecture.

The design is suitable for:

- agricultural lenders,
- portfolio-based bank insurance programs,
- input finance schemes,
- aggregators and contract-farming networks,
- insurers and reinsurers,
- donor-backed agricultural resilience facilities.

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3. Product Identity and Core Philosophy

3.1 Product Name

Verdure Trace Area Yield Index Insurance-plus or simply *AYII+*

3.2 Product Type

This is an **area-yield index insurance product enhanced by weather and NDVI**, not a pure weather index product and not a pure vegetation index product.

3.3 Core philosophy

The central philosophy is:

- **Weather tells us whether an insured peril occurred.**
- **NDVI tells us whether the crop responded negatively during the insured period.**
- **Yield logic tells us how much seasonal production value was ultimately lost.**

This preserves the AYII foundation. The reference document explicitly defines AYII compensation as being triggered when actual area yield falls below benchmark yield and states that benchmark yield is set as a proportion of historical average yield.

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4. Why Verdure Trace Needs a Hybrid AYII

The reference AYII design recognizes that product development starts from cleaned yield data and proxy data such as weather and satellite data, and that a major challenge is

reviewing the accuracy of collected data and validating datasets with proxy data. That is exactly the gap Verdure Trace is solving.

Traditional AYII can be weakened by:

- unreliable yield submissions,
- delayed or incomplete settlement data,
- hidden volatility in yield data,
- overstatement of expected yields,
- basis risk between farm losses and area averages.

The reference document itself warns of hidden volatility in yield data and possible errors in village-level reported yields, and notes that lower indemnity levels may be needed in early implementation because of such uncertainty.

The hybrid design addresses this by adding:

- NDVI-based yield validation,
- independent weather-based burn-cost simulation,
- modelled yield estimation,
- cumulative in-season value-at-risk logic.

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5. Scope and Insurance Unit

5.1 Recommended insured unit

The recommended insured unit on Verdure Trace is the **Geo Cluster**.

This is preferred over a broad AEZ because it better balances:

- homogeneity,
- operational practicality,
- granularity,
- pricing precision,
- and basis-risk control.

The reference design notes that settlement granularity should be chosen based on historical dataset availability and that its pilot chose village-level settlement.

5.2 Verdure Trace hierarchy

The product should be configured through:

Project → AEZ → Geo Cluster → Crop → Variety → Season

5.3 Initial crop rollout

Recommended pilot crop:

- maize

Subsequent expansion:

- rice
- sorghum
- soybean
- wheat
- selected horticulture where remote sensing performance is adequate

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6. Covered Risks and Insurance Period

6.1 Covered perils

Phase 1 should cover:

- drought,
- excess rainfall / flood stress,
- heat stress.

The reference design covers floods, drought, and plant destruction organisms, while prescribing exclusions for wider natural disasters such as earthquakes, tsunamis, volcanic eruptions, and wildfires.

For Verdure Trace, pest and disease risk should initially be treated as:

- a monitoring overlay,
- an explanatory layer,
- or a future extension,

  rather than an automated primary payout trigger unless there is strong calibration evidence.

6.2 Insurance period

The insurance period should cover one planting season, up to six months. The reference design uses one planting season with a maximum of six months and states that cover starts 30 days after planting.

For Verdure Trace, recommended configuration is:

- **Cover start:** planting date or planting date plus waiting period,
- **Cover end:** harvest date proxy or defined seasonal end,
- **Monitoring frequency:** daily weather, dekadal or biweekly NDVI.

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7. Data Framework

7.1 Historical yield data

The reference document states that a minimum of 10 years of yield data is required for product design and that yield data must be checked for anomalies in recording and entry. It also explains that yield data is used primarily to calculate average yield values.

Verdure Trace rule

- minimum acceptable: 7 years,
- preferred: 10 years,
- stronger pricing standard: 10+ years with anomaly review and detrending.

7.2 Weather data

The reference design lists weather data including rainfall, temperature, and wind, ideally over 30 years, for understanding probable maximum loss and evaluating utility in dispute resolution.

Verdure Trace rule

Use:

- rainfall,
- temperature,
- dry spell statistics,
- extreme rainfall windows,
- optional soil moisture,
- optional evapotranspiration,
- optional flood extent or waterlogging indicators.

Recommended history:

- 20 to 30 years for catastrophe analysis,
- shorter periods may be used for preliminary pilots.

7.3 Remote sensing data

The reference document explicitly lists NDVI, EVI, VCI, and flood or drought extent as useful remote sensing inputs for validating yield data and understanding variation in the data.

Verdure Trace rule

- NDVI history: minimum 7 years, preferred 10 years,
- cloud masking and smoothing required,
- crop-mask enforcement required,
- alternative vegetation or moisture indices can be used later where NDVI saturates.

7.4 Other required data

- area sown,
- area insured,
- crop calendar,
- planting window,
- variety with normal yield potential,
- irrigation classification,
- historical premiums and losses where available.

The reference document also identifies area sown, variety data, area insured, and past premium/loss information as useful for risk evaluation and product design.

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8. Product Philosophy on Yield

8.1 Declared yield is provisional

Client-submitted historical yield should be treated as a provisional underwriting input, not automatic truth.

8.2 NDVI is not the legal definition of yield

NDVI should not be presented as identical to harvested yield. It should instead be positioned as:

- a crop-condition proxy,
- a validation layer,
- an explanatory variable in yield modelling,
- an in-season stress indicator.

8.3 Yield credibility framework

Verdure Trace should score submitted yield data using:

- NDVI consistency with yield trend,

- weather consistency with declared outcomes,
- data continuity,
- outlier detection,
- agronomic plausibility.

This is aligned with the reference document's emphasis on checking anomalies in yield data and validating datasets with proxy data.

8.4 Benchmark hierarchy

Each pricing run should calculate:

Declared Benchmark Yield

Derived from client-submitted history.

NDVI-Validated Benchmark Yield

Declared benchmark adjusted using historical NDVI consistency.

Insurable Benchmark Yield

Final benchmark accepted after:

- NDVI validation,
- weather backtesting,
- burn-cost simulation,
- profitability and tail-risk constraints.

A practical rule is:

BY_{insurable} = \min(BY_{declared}, BY_{NDVI-adjusted}, BY_{risk-acceptable})

This reflects the desired Verdure Trace underwriting logic: the system can reduce an unrealistic benchmark to an insurable level supported by its own evidence layers.

9. Benchmark Yield Methodology

The reference AYII framework states that the trigger in AYII is benchmark yield shown as a percentage of the historical average yield, and in the cited pilot benchmark yield was set at 85 percent of the average yield at settlement-unit level.

9.1 Base formula

$$BY = c \times \bar{Y}_{hist}$$

Where:

- BY = benchmark yield,
- c = indemnity or coverage proportion,
- $\bar{Y}_{hist}$ = detrended historical average area yield.

9.2 Recommended coverage level

For pilot deployment:

- 80% to 85% benchmark level is prudent.

This is consistent with the reference design and also with its caution that lower indemnity levels may be recommended when hidden volatility or yield data uncertainty exists.

9.3 Detrending

Historical yield should be detrended where:

- seed quality has improved,
- fertilizer intensity changed,
- agronomic support improved,
- irrigation status changed.

9.4 Variety potential

The reference design includes variety data with normal yield potential as part of validation and explanation of losses.

Verdure Trace should use variety potential as:

- a ceiling,
- a plausibility bound,
- a validation reference,

  not as the default insured benchmark unless supported by historical and modelled evidence.

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10. NDVI Methodology

10.1 Functional role of NDVI

NDVI in Verdure Trace serves four functions:

1. validate submitted historical yield,
2. extend historical crop-condition insight,
3. monitor crop condition during the season,
4. support modeled area-yield estimation.

10.2 Historical NDVI seasonal baseline

For each Geo Cluster, crop, variety, and planting window, Verdure Trace should build a seasonal NDVI baseline:

$$NDVI^{base}_{w}$$

Where w is time since planting.

Store:

- median trajectory,
- 25th percentile,
- 10th percentile,
- anomaly band by growth stage.

10.3 NDVI stress score

A normalized stress score can be defined as:

$$N_s = \frac{NDVI^{base}_{s} - NDVI^{obs}_{s}}{NDVI^{base}_{s}}$$

Where:

- N_s = NDVI stress at stage s,
- $NDVI^{base}_{s}$ = expected NDVI at stage s,
- $NDVI^{obs}_{s}$ = observed NDVI at stage s.

Cap and normalize this score between 0 and 1.

10.4 NDVI processing controls

Required controls:

- cloud masking,
- temporal compositing,
- smoothing,
- crop mask enforcement,
- valid-pixel thresholds,
- outlier filtering.

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11. Weather Methodology

11.1 Role of weather

Weather is the peril-detection layer. It should identify:

- rainfall deficit,
- prolonged dry spell,
- excess rainfall,
- flood-compatible rainfall episodes,
- heat stress.

11.2 Weather variables

Recommended metrics:

- cumulative rainfall by rolling window,
- rainfall anomaly against local climatology,
- consecutive dry days,
- 3-day and 5-day extreme rainfall,
- days above heat threshold,
- optional soil moisture deviation,
- optional flood extent linkage.

11.3 Weather stress score

W_s = f(RainDeficit_s, DrySpell_s, ExcessRain_s, Heat_s, Flood_s)

Scale W_s from 0 to 1 after calibration.

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12. Trigger Architecture

12.1 Dual-confirmation eligibility trigger

A payout becomes eligible when:

Condition 1

A weather peril threshold is breached.

Condition 2

NDVI confirms crop deterioration during the same or expected lagged stage window.

This matches the design intention that NDVI should monitor vegetation during the season as a proxy for yield and that NDVI decline within the insured period should indicate likely bad yield conditions.

12.2 Why dual confirmation is superior

This reduces false positives from:

- bad weather without crop impact,
- NDVI noise unrelated to peril,
- cloud contamination,
- non-covered causes of vegetation decline.

12.3 Stage-specific calibration

Thresholds should differ by crop stage:

- establishment,
- vegetative,
- flowering,
- grain fill / maturity.

The same NDVI or rainfall deviation should not be treated equally across all stages.

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13. Settlement Architecture

The reference framework defines AYII compensation as a function of the relationship between actual yield and benchmark yield.

13.1 Preferred settlement approach

Verdure Trace will settle on:

- actual area yield where reliable, or
- modelled area yield where actual area yield is delayed, incomplete, or not sufficiently credible.

13.2 Modelled area yield

A practical first model is:

$$\hat{Y}_{area} = \alpha + \beta_1 NDVI_{peak} + \beta_2 NDVI_{integral} + \beta_3 Rain_{cum} + \beta_4 DrySpell + \beta_5 HeatDays + \epsilon$$

Later this can be improved with machine-learning or hierarchical models.

13.3 Final payout rate

$$Payout\ Rate = \max\left(0,\frac{BY_{insurable}-\hat{Y}_{area}}{BY_{insurable}}\right)$$

13.4 Final payout amount

$$Claim = SI \times Area \times Payout\ Rate$$

Where:

- SI = sum insured per hectare,
- Area = insured hectares.

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14. Revised Cumulative In-Season Payout Layer

This section is fully revised to reflect the new feedback.

14.1 Concept

Where Verdure Trace offers in-season payouts, stage weights must represent **incremental crop value accumulation**, but the payout at any point in the season must be assessed on a **cumulative exposure basis**.

This means a loss at a later stage reflects not just that stage's value, but the value already accumulated in all prior stages.

14.2 Example stage schedule

Illustrative incremental stage weights:

- Establishment: 20%
- Vegetative: 25%
- Flowering: 35%
- Grain fill / maturity: 20%

These sum to 100 percent of the seasonal insured value.

14.3 Cumulative stage exposure

The cumulative exposure profile is therefore:

- End of Establishment: 20%
- End of Vegetative: 45%
- End of Flowering: 80%

- End of Grain fill / maturity: 100%

This means:

- a catastrophic loss at establishment can justify up to 20%,
- a catastrophic loss at vegetative stage can justify up to 45%,
- a catastrophic loss at flowering can justify up to 80%,
- a catastrophic loss at maturity can justify up to 100%.

That is the correct agronomic and insurance logic because crop value at risk accumulates as the crop develops.

14.4 Formal notation

Let:

- w_s = incremental stage weight,
- C_s = cumulative exposure up to stage s.

Then:

$$C_s = \sum_{i=1}^{s} w_i$$

If:

- $w_1 = 0.20$,
- $w_2 = 0.25$,
- $w_3 = 0.35$,
- $w_4 = 0.20$,

then:

- $C_1 = 0.20$,
- $C_2 = 0.45$,
- $C_3 = 0.80$,
- $C_4 = 1.00$.

14.5 Stage severity

Stage loss severity should be estimated from the hybrid trigger evidence:

$$L_s = \gamma W_s + (1-\gamma)N_s$$

Where:

- W_s = weather stress score at stage s,

- N_s = NDVI stress score at stage s,
- γ = weighting parameter.

L_s should be bounded between 0 and 1.

14.6 Cumulative stage payout

If a covered loss is assessed at stage s, the target cumulative payout is:

$$TargetCumulativePayout_s = SI \times C_s \times L_s$$

So if:

- seasonal sum insured is 100,
- cumulative exposure at vegetative stage is 45%,
- stage severity is total loss, $L_s = 1$,

then target cumulative payout is 45.

14.7 Incremental payment rule

To avoid double counting, Verdure Trace must track what has already been paid. If prior payments exist, the next payment is:

$$Incremental\ Payout_t = \max(0,\ TargetCumulativePayout_t - PaidToDate_{t-1})$$

This means if:

- 10% was already paid at establishment,
- later vegetative-stage evidence justifies 45% cumulative payout,

the additional payment is 35%, not another full 45%.

14.8 Practical policy rule

At every stage, Verdure Trace should maintain:

- stage increment weight,
- cumulative exposure cap,
- paid-to-date percentage,
- current target cumulative payout.

14.9 Final reconciliation

At end of season:

- the cumulative in-season payouts should be reconciled against the final AYII settlement,
- any underpayment can be topped up,
- any overpayment should be controlled contractually through conservative in-season payout rules or treated as final subject to product wording.

14.10 Positioning of this layer

This cumulative in-season layer is best treated as:

- an optional enhancement,
- an early-liquidity mechanism,
- a staged advance against final end-of-season liability.

The base legal identity of the product should still remain AYII.

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15. Pricing Framework

The reference design states that ratemaking follows the determination of indemnity level, benchmark yield, sum insured, and expected historical payoffs, and that premium calculation is a key step after those choices are made.

15.1 Gross premium formula

$$Gross\ Premium = EL + VL + CL + Admin + Distribution + Margin$$

Where:

- EL = expected loss,
- VL = volatility loading,
- CL = catastrophe loading.

15.2 Expected loss

Expected loss comes from historical backtesting of the hybrid trigger and settlement model:

$$EL = \frac{1}{n}\sum_{t=1}^{n} Payout_t$$

15.3 Frequency and severity

Pure premium can also be expressed as:

$$Pure\ Premium = Frequency \times Severity$$

Where:

- frequency = proportion of seasons with payout,
- severity = average payout conditional on payout.

15.4 Catastrophic loading

The reference design uses long weather series to understand probable maximum loss. Verdure Trace should therefore use long weather histories to price:

- severe drought years,
- extreme rainfall / flood years,
- cluster tail loss.

15.5 Commercial guardrails

Set underwriting constraints such as:

- maximum acceptable expected loss,
- maximum payout frequency,
- maximum tail exposure,
- minimum premium floor,
- decline rule for uninsurable clusters.

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16. Underwriting Logic

16.1 Objective

Verdure Trace should determine the highest defensible insurable benchmark and corresponding trigger structure that still satisfies:

- profitability,
- data credibility,
- payout stability,
- catastrophe tolerance.

16.2 Optimization constraints

Use constraints such as:

- minimum NDVI-yield consistency,
- minimum weather-yield consistency,
- maximum expected loss,
- maximum payout frequency,
- acceptable tail loss,

- minimum data quality.

16.3 Actions if risk is too high

Verdure Trace may:

- lower benchmark yield,
- lower indemnity level,
- reduce sum insured,
- narrow peril scope,
- increase premium,
- decline the risk.

This formalizes the core design intent that the platform should not insure yield levels that its own evidence layers cannot support.

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17. Basis Risk Management

The reference design explicitly notes that basis risk is inherent in index products and that farmers may incur production losses without receiving payouts.

17.1 Key sources of basis risk

- cluster too large,
- mixed pixels,
- poor crop masks,
- planting-date mismatch,
- irrigation inside rainfed clusters,
- NDVI saturation,
- poor historical yield quality,
- settlement data errors.

17.2 Controls

Verdure Trace should use:

- Geo Cluster settlement,
- crop masks,
- planting window logic,
- irrigation flags,
- confidence scoring,
- manual review for outliers,
- optional field validation samples,
- crop cutting experiments or equivalent sampled evidence for calibration and audit.

The reference design uses crop cutting experiments to obtain area yield in standard AYII.

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18. Governance, Explainability, and Defensibility

To be accepted by banks, insurers, reinsurers, and technical reviewers, every pricing and claim output must be explainable.

18.1 Pricing output should include

- benchmark source,
- NDVI validation result,
- weather backtest summary,
- payout frequency,
- expected loss,
- catastrophe loading,
- final premium,
- confidence score.

18.2 Claim output should include

- peril timeline,
- weather breach evidence,
- NDVI anomaly curve,
- cumulative stage exposure where relevant,
- modeled yield estimate,
- benchmark yield,
- payout rate,
- paid-to-date amount,
- final incremental payout.

18.3 Audit pathway

For disputes or unusual claims, support:

- crop cutting checks,
- extension officer review,
- procurement or warehouse data cross-checks,
- third-party agronomic validation.

The reference design explicitly mentions remote sensing and weather also being assessed for dispute-resolution utility.

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19. Verdure Trace System Design Requirements

19.1 Proposed module flow

Tab 1: Selection Hierarchy

- Project
- AEZ
- Geo Cluster
- Crop
- Variety
- Season

Tab 2: Data Configuration

- cover start and end,
- yield source,
- weather source,
- NDVI source,
- years used,
- detrending toggle,
- crop calendar.

Tab 3: Yield Validation

- declared yields,
- NDVI correlation view,
- weather consistency check,
- adjusted benchmark recommendation,
- confidence score.

Tab 4: Trigger Engine

- peril selection,
- stage-specific weather thresholds,
- stage-specific NDVI thresholds,
- lag and persistence rules,
- AND logic for dual confirmation.

Tab 5: Yield Model

- benchmark yield,
- modeled area yield,
- backtest diagnostics,
- basis-risk indicator.

Tab 6: Pricing Engine

- expected loss,
- frequency,
- severity,
- volatility loading,
- catastrophe loading,
- gross premium.

Tab 7: In-Season Payout Engine

- stage increment weights,
- cumulative exposure curve,
- stage severity,
- paid-to-date tracker,
- incremental payout calculator.

Tab 8: Saved Pricing

- scenario versions,
- underwriter notes,
- approval workflow.

Tab 9: Monitoring

- weather signal dashboard,
- NDVI anomaly dashboard,
- stage stress indicators,
- provisional payout estimate,
- reserve estimate.

Tab 10: Reports

- product sheet,
- underwriting memo,
- pricing report,
- claim decision note,
- audit pack.

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20. Recommended Pilot Rollout

20.1 Phase 1: Shadow testing

- run product on historical data,

- compare to observed outcomes,
- calibrate NDVI-yield and weather-yield relationships,
- refine cumulative stage schedule.

20.2 Phase 2: Controlled live pilot

- maize only,
- selected Geo Clusters,
- drought and excess rainfall only,
- manual review of major payouts,
- conservative in-season cumulative payouts.

20.3 Phase 3: Scale-up

- more crops,
- more geographies,
- insurer and bank dashboards,
- reinsurance reporting,
- stronger automation.

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21. Recommended Policy Positioning Language

Use this wording:

Verdure Trace *AYII+* is an area-yield insurance product that combines historical yield evidence, weather analytics, and satellite-derived vegetation monitoring to assess seasonal crop performance. Weather identifies the occurrence of insured perils, NDVI validates crop response during the season, and payouts are determined by the shortfall of observed or modelled area yield relative to a benchmark yield for the insured cluster. Where enabled, cumulative in-season payouts may be made based on the proportion of crop value accumulated by each stage of growth, subject to final seasonal reconciliation.

This wording is strong because it:

- preserves AYII identity,
- positions NDVI correctly,
- allows the cumulative stage feature,
- remains defensible.

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22. Final Recommendation

The fully revised recommended design for Verdure Trace is:

a *AYII+* product in which declared yield is treated as provisional, NDVI is used to validate and constrain yield assumptions, weather defines peril occurrence and burn-cost structure, final settlement is based on observed or modelled area-yield shortfall, and optional in-season payouts follow a cumulative crop-value schedule across growth stages.

That is the most coherent, comprehensive, and defensible form of the concept. The AYII reference supports the core structure by grounding the product in benchmark yield and actual area yield, requiring historical yield data, recognizing weather for PML and pricing support, and recognizing remote sensing for yield validation and dispute utility.